

BRANCH PATH: SPARSE SENSOR MEMORY

INTERACTION

OUTPUT AND LOSS

Observations

sensor values,
positions
+ time

Fourier embedding

1

CfC Memory

token attention,
causal scan dt

K,V

2

Cross-Attention Readout

isotropic
distance bias

Branch Basis

inner product

Field

u, v, p

autograd

3

Loss

AL continuity
constraint,
sensor MSE + NS
residual, GradNorm

optimize

TRUNK PATH: QUERY FEATURE

Query

(x, y, t_q, c)

Trunk Feature

Fourier +
temporal anchor,
dt_to_query

Q

Trunk Basis

update θ

SOAP + Schedule-Free

preconditioned
updates